

Exercise 3 SQL CASE statements

Q1 The Products Table

```
SELECT product-id, product-name
CASE
  WHEN price > 1000 THEN 'expensive'
  WHEN price is BETWEEN 700 and 1000 THEN 'midrange'
  ELSE 'Budget'
END AS Price category
FROM Products;
```

Product name	Price	Price category
laptop	1200.00	expensive
phone	800.00	Mid-range
keyboard	45.00	Budget
Monitor	300.00	Mid range
Mouse	25.00	Budget

Q2: Orders Table

```
SELECT customer-name,
       amount
```

```
CASE
  WHEN amount ≥ 1000 THEN 'high value'
  WHEN amount is BETWEEN 500 AND 999.99 THEN 'medium value'
  ELSE 'Low value'
END AS 'order value category'
FROM orders;
```

Customer name	amount	order value category
Alice	150.00	low value
Bob	560.00	Medium value
Charlie	999.99	Medium value
Diana	45.50	low value
Ethan	1200.00	High value

Q3
→

```
SELECT emp_name,
       department,
       salary
```

CASE

~~WHEN~~ IF

```
WHEN department = 'IT' AND salary > 8000 THEN 'Senior IT'
WHEN department = 'HR' AND salary > 50000 THEN 'Experienced HR'
ELSE 'Staff'
END AS Position level
FROM Employee;
```

Emp_name	department	Salary	Position
Jain	IT	85000	Senior IT
Saral	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
TOM	HR	55.000	Staff

Q4 Student Table
 SELECT student_name,
 Score

CASE

WHEN score ≥ 90 THEN 'A'

WHEN score BETWEEN 80 AND 90 THEN 'B'

WHEN score BETWEEN 70 AND 80 THEN 'C'

WHEN score BETWEEN 60 AND 69 THEN 'D'

WHEN score > 60 THEN 'F'

END AS grade
 FROM student;

Student_name	Score	Grade
Anna	92	A
Ben	76	C
Core	59	F
David	83	B
Bila	68	D

Q5 Delivery Table

SELECT delivery_id,
 delivery_time_minutes

CASE

WHEN delivery time minutes ≤ 30 mins THEN 'Fast'

WHEN delivery time minutes BETWEEN 31 AND 60 THEN 'On time'

WHEN delivery time minutes > 60 THEN 'late'

END AS Performance

FROM delivery;

delivery_id	delivery_time - minutes	Performance
1	45	On time
2	80	late
3	30	Fast
4	65	late
5	100	late

Q6 Tickets Table

```
SELECT issue_type,
       priority
```

```
CASE
  WHEN priority = 3 THEN 'High'
  WHEN priority = 2 THEN 'Medium'
  WHEN priority = 1 THEN 'Low'
```

```
END AS priority_label
```

```
FROM tickets;
```

Result

Issue type	Priority	Priority level
login issue	1	Low
Server down	3	High
Slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

Q7

```
SELECT student_id,
       days present, total days
```

$$\frac{48}{50} \times 100 = 96$$

$$(\text{days present} \times 100.0) / \text{total days}$$

```
FROM tickets;
```

```
CASE WHEN attendance percentage ≥ 90 THEN 'Excellent'
```

```
  WHEN attendance percentage BETWEEN 75 AND 89
    THEN 'Good'
```

```
  WHEN attendance percentage < 75 THEN 'Needs Improvement'
```

```
END AS attendance status
```

```
FROM tickets;
```

90%

60%

96%

50%

100%

Results Attendance Table

Student_id	attendance - percentage	attendance-status
1	90	Excellent
2	60	Needs Improvement
3	96	Excellent
4	50	Needs Improvement
5	100	Excellent

Q8 ^{Inventory} Products Table

```
SELECT product_id,
        stock_qty,
```

```
CASE
  WHEN stock_qty = 0 THEN 'Out of Stock'
  WHEN stock_qty BETWEEN 1 AND 5 THEN 'Low Stock'
  WHEN stock > 5 THEN 'In Stock'
```

```
END AS Stock Status
```

```
FROM products-inventory;
```

Results

Product_id	stock_qty	stock-status
1	5	Low Stock
2	0	Out of Stock
3	25	In Stock
4	10	In Stock
5	3	Low Stock

Q9 The Classes Table

```
SELECT subject,
        enrolled_students
```

```
CASE
```

```
WHEN enrolled_students  $\geq$  25 THEN 'large'
```

```
WHEN enrolled_students BETWEEN 10 AND 24 THEN 'Medium'
```

```
WHEN enrolled_students < 10 THEN 'small'
```

```
END AS class_size_category
```

```
FROM classes;
```

results

subject	enrolled_students	class_size_category
Maths	30	large
English	25	large
Science	15	Medium
Art	5	small
History	20	Medium

Q10 The Payments Table.

```
SELECT payment_id, amount, payment_method
```

```
CASE
```

```
WHEN payment_method = 'Cash' AND amount  $\geq$  200 THEN  
    'Eligible for discount'
```

```
ELSE 'Not Eligible'
```

```
END AS discount_eligibility
```

```
FROM the_payments;
```

payment_id	amount	payment_method	discount_eligibility
1	50.00	Card	Not eligible
2	200.00	Cash	Eligible for discount
3	150.00	Card	Not eligible
4	75.00	Paypal	Not eligible
5	300.00	Cash	Eligible for discount